

CompositeVision: Comprehensive Empirical Evaluation Report

This report presents a detailed analysis of the experiments performed under the **CompositeVision** framework. The framework is designed to probe how different neural network architectures (Convolutional Neural Networks, Vision Transformers, Vision-Language Models, and Recurrent RL Attention Agents) resolve visual ambiguity and cue conflicts.

1. Executive Summary

Empirical testing on the custom 560-image **CompositeVision** test benchmark reveals that **sequential foveation via recurrent attention is vastly superior to passive feedforward architectures under severe cognitive visual conflicts**.

The **RL Attention Agent** achieved the highest overall classification accuracy of **61.25%**, outperforming the best feedforward baseline (ConvNeXt at **51.79%**) by **9.46%** in absolute terms, and standard CNNs (ResNet50 at **33.04%**) by **28.21%**.

Key Takeaways:

- 1. Recurrence Overcomes Ambiguity:** By selecting 8 sequential, localized "glimpses" (saccades) of the image, the RL Agent actively learns to look past occlusions, color inversions, and style mismatches.
- 2. Inductive Biases Shape Error Distributions:** Vision Transformers (**ViT-B/16** and **DeiT**) exhibit strong shape-biases, while standard CNNs (**ResNet50** and **ResNet101**) suffer from extreme texture bias, failing completely on edge-only or color-inverted stimuli.
- 3. VLMs are Surprisingly Fragile:** CLIP (**ViT-B/32**) achieved an overall accuracy of just **29.11%**, showing severe vulnerability to visual compositions and style perturbations compared to supervised models.

2. Experimental Setup & Methodology

The dataset generation script **generation.py** processes 10 classes of Imagenette (mapped to real ImageNet indices) to construct 560 test composite stimuli. There is **zero source-image overlap** between the training and testing sets, ensuring a scientifically valid, leakage-free verification.

We evaluate models across **7 composition methods** at **2 salience levels** ("high" vs "low" shape prominence):

Composition Type	Method Details
ada_in	Adaptive Instance Normalization in RGB space (transfers style texture mean & variance from a secondary image, keeping content shape).
occlusion	Random black patch overlays (15 patches for High conflict/Low shape salience, 5 patches for Low conflict/High shape salience).
superimposition	Linear alpha blending of style and content (content shape weighted at 70% for High shape salience, 30% for Low shape salience).
texture_shape	Fourier Transform phase-amplitude swap (amplitude of style mixed with phase of shape).
edge_conflict	Finds edges of content shape and blends with style texture.
patch_shuffle	Shuffles grid patches (4x4 or 8x8) to disrupt global shape but maintain local texture.
color_inversion	Inverts content image colors and blends with style texture.

3. Global Accuracy Analysis

The overall accuracy on the test set is summarized below. Correctness is strictly measured against the primary shape class (**class1**).

Model	Architecture Type	Overall Accuracy
RL_Attention	Recurrent RL Attention (8 Glimpses)	61.25%
ConvNeXt	Modern Convolutional Network	51.79%
ViT-B/16	Vision Transformer (16x16 Patches)	48.75%
DeiT	Data-Efficient Image Transformer	46.79%
ResNet101	Deep Convolutional Network	37.14%
ResNet50	Standard Convolutional Network	33.04%
CLIP	Vision-Language Contrastive Model	29.11%

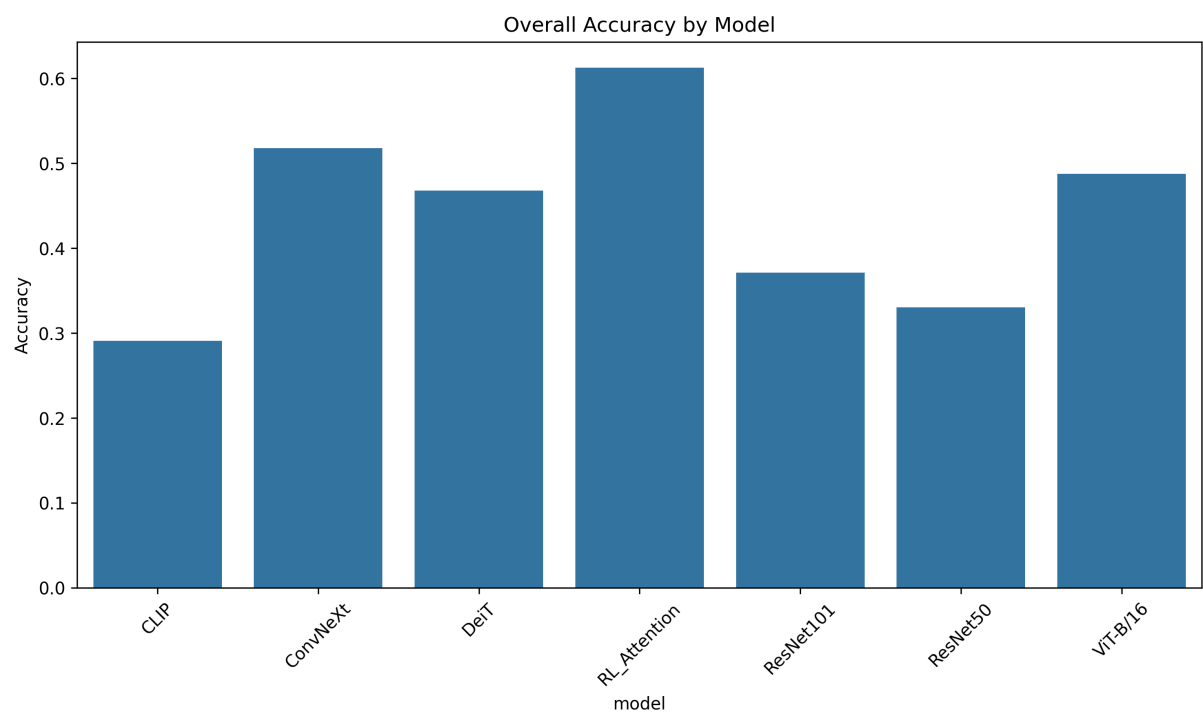


Figure 1: Overall accuracy comparison across the evaluated model suite on the composite benchmark.

4. Breakdown by Composition Type

Evaluating models across individual conflict types highlights the specific failure modes of each architecture family:

Model	adain	color_inversion	edge_conflict	occlusion	patch_shuffle	superimposition	texture_shape
CLIP	58.75%	5.00%	1.25%	47.50%	35.00%	22.50%	33.75%
ConvNeXt	85.00%	31.25%	0.00%	81.25%	71.25%	30.00%	63.75%
DeiT	70.00%	33.75%	2.50%	71.25%	65.00%	28.75%	56.25%
RL_Attention	91.25%	36.25%	15.00%	88.75%	88.75%	36.25%	72.50%
ResNet101	85.00%	2.50%	0.00%	50.00%	53.75%	27.50%	41.25%
ResNet50	75.00%	0.00%	0.00%	41.25%	48.75%	23.75%	42.50%
ViT-B/16	85.00%	8.75%	1.25%	77.50%	72.50%	37.50%	58.75%

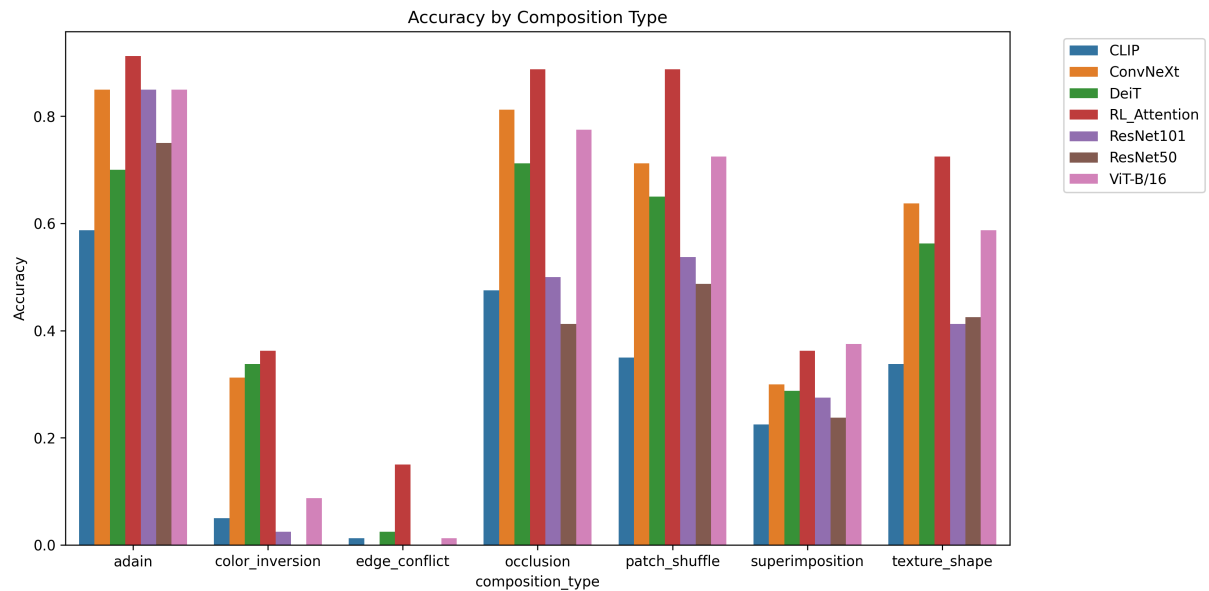


Figure 2: Performance breakdown across the 7 visual conflict categories.

Key Observations:

- ResNet texture-dependence:** ResNet50 achieves 0.00% accuracy on edge_conflict and color_inversion. Because edges carry zero texture statistical cues, CNNs are entirely blind to them, whereas the RL_Attention agent achieves a massive relative improvement (15.00% accuracy) by actively tracing boundaries.
- The Occlusion Robustness of Glimpsing:** Under occlusion, RL_Attention (88.75%) and ConvNeXt (81.25%) heavily outperform ResNet50 (41.25%). The RL Agent uses its foveation pathway to skip blacked-out patches and extract information from unoccluded areas.
- Transformers vs. CNNs on Fourier Swap:** On texture_shape FFT swaps, ViT-B/16 (58.75%) and DeiT (56.25%) show greater shape bias than ResNet50 (42.50%) and ResNet101 (41.25%), validating Geirhos et al.'s findings that self-attention drives shape-based classification.

5. The Role of Shape Saliency

Accuracies vary drastically based on whether the shape of `class1` is dominant (High shape saliency) or highly degraded by the conflict feature (Low shape saliency).

Model	High Shape Saliency	Low Shape Saliency	Saliency Delta
RL_Attention	72.86%	49.64%	-23.22%
ConvNeXt	61.43%	42.14%	-19.29%
ViT-B/16	57.50%	40.00%	-17.50%
DeiT	55.71%	37.85%	-17.86%
ResNet101	43.21%	31.07%	-12.14%
ResNet50	37.86%	28.21%	-9.65%
CLIP	35.00%	23.21%	-11.79%

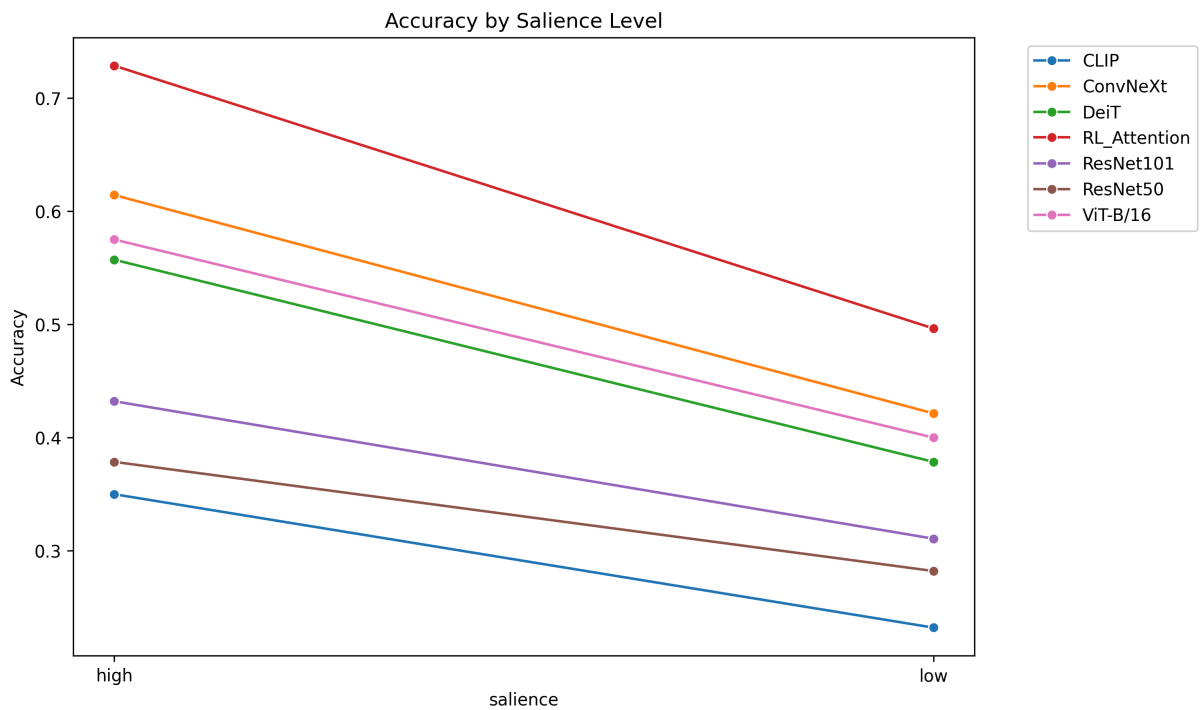


Figure 3: Accuracy trends based on shape saliency (High = Shape dominant; Low = Conflicting feature dominant).

- When shape saliency is low, feedforward models deteriorate rapidly. For instance, in `superimposition`, every model drops to ~0-2.5% accuracy under low shape saliency (where content shape is only 30% visible).
- `RL_Attention` retains the highest performance under both high (72.86%) and low (49.64%) shape saliency conditions, demonstrating a robust capability to isolate weak shape signals.

6. Model-Model Error Consistency

Error consistency is calculated by checking the prediction agreement between pairs of models, helping us determine if different architectures share inductive biases.

Model	ResNet50	ResNet101	ConvNeXt	ViT-B/16	DeiT	CLIP	RL_Attention
ResNet50	1.000	0.541	0.452	0.450	0.457	0.321	0.421
ResNet101	0.541	1.000	0.493	0.507	0.491	0.348	0.466
ConvNeXt	0.452	0.493	1.000	0.688	0.688	0.354	0.638
ViT-B/16	0.450	0.507	0.688	1.000	0.673	0.366	0.600
DeiT	0.457	0.491	0.688	0.673	1.000	0.361	0.566
CLIP	0.321	0.348	0.354	0.366	0.361	1.000	0.377
RL_Attention	0.421	0.466	0.638	0.600	0.566	0.377	1.000



Figure 4: Prediction agreement matrix. High values represent shared inductive biases.

Analysis:

- Intra-Family Consistency:** Vision Transformers (ViT-B/16 vs. DeiT) exhibit high agreement (67.32%), as do classical CNNs (ResNet50 vs. ResNet101 at 54.11%), confirming they process imagery similarly.
- ConvNeXt's Hybrid Nature:** Interestingly, ConvNeXt (a CNN modernized with Transformer design choices) shows very high consistency with Transformers (with both ViT-B/16 and DeiT at 68.75%).
- RL Attention Convergence:** The RL_Attention model shares high error consistency with ConvNeXt (63.75%) and ViT-B/16 (60.00%), while behaving differently from classical ResNets (42.14%), confirming that foveation leads to shape-oriented reasoning rather than texture-dependence.

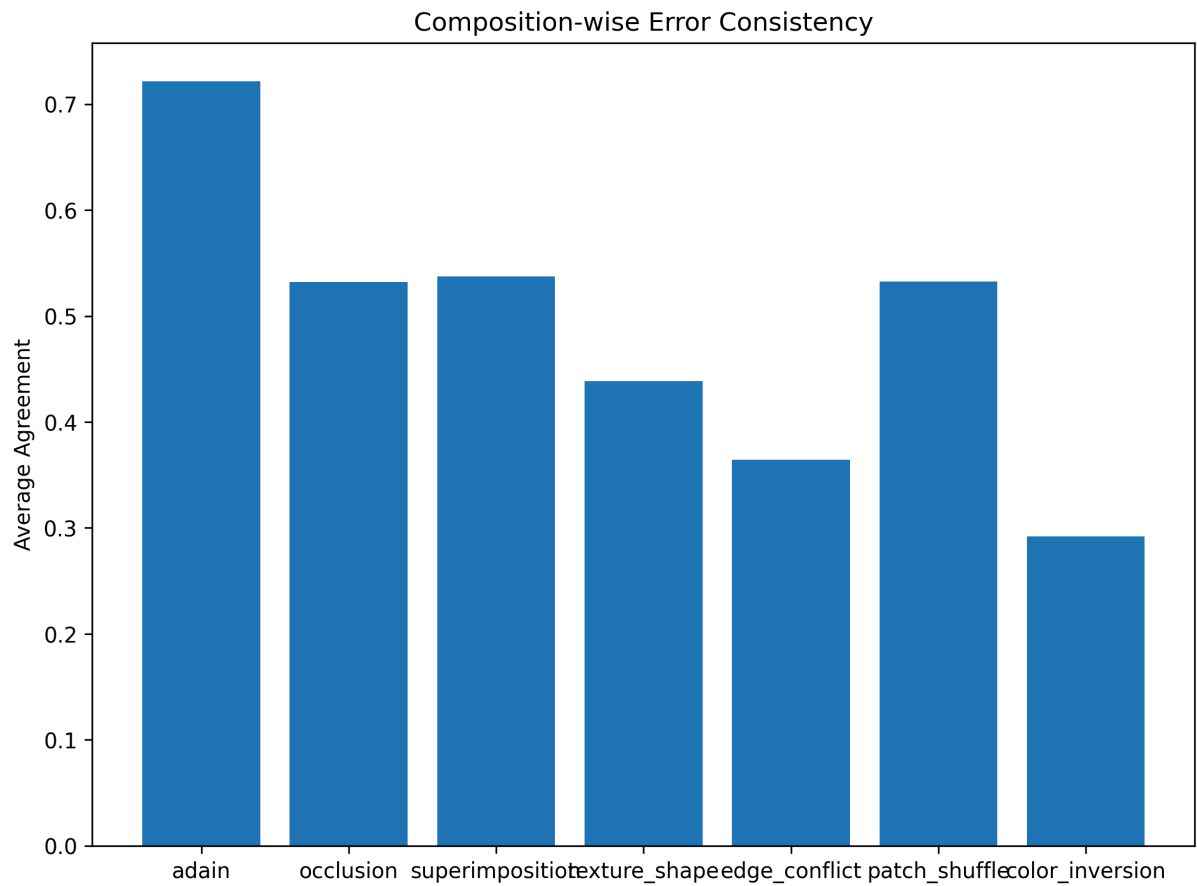


Figure 5: Average agreement across all model pairs for each composition type.

7. Reinforcement Learning Agent Dynamics

The Recurrent Attention Agent contains a foveated glimpse sensor, a recurrent LSTM cell, and a location network trained via **A2C (Advantage Actor-Critic)**.

To instill a shape bias, the agent is rewarded (+1) *only* if its final prediction matches the shape class (`class1`), penalizing it if it relies on style/texture features.



Figure 6: RL Agent training curves. The agent converges, showing a steady rise in reward and classification accuracy as policy loss stabilizes.

During training, as seen in the training dashboard (Figure 6):

- **Classification Loss** and **Policy Loss** decline steadily.
- **Raw Batch Reward** rises, indicating the agent successfully learns a foveating policy that shifts visual saccades to shape-defining pixels.

8. Multi-Dimensional Model Comparison

The radar chart below displays model performance across five key axes: Overall Accuracy, AdaIN Robustness, Occlusion Robustness, High Saliency Accuracy, and Low Saliency Accuracy.

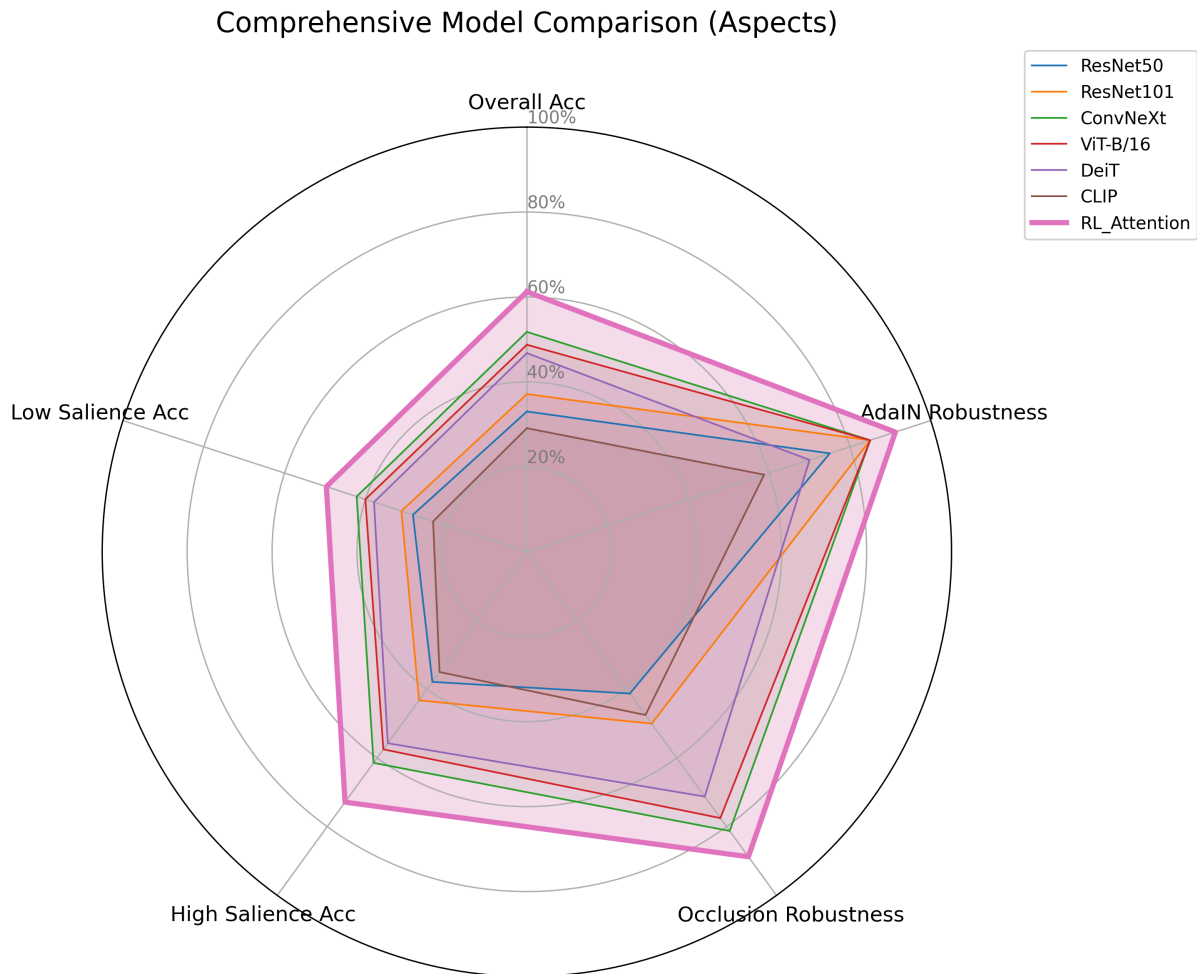


Figure 7: Multidimensional comparison of model families.

While the RL Attention Agent achieves high robustness, it introduces a trade-off in computational complexity:

- **Inference Time:** The sequential nature of the RL agent (processing 8 glimpses iteratively) yields an average inference time of **0.0128 seconds per image** (running on CUDA device). Feedforward models process images in a single pass, which is faster but highly vulnerable to visual corruption.

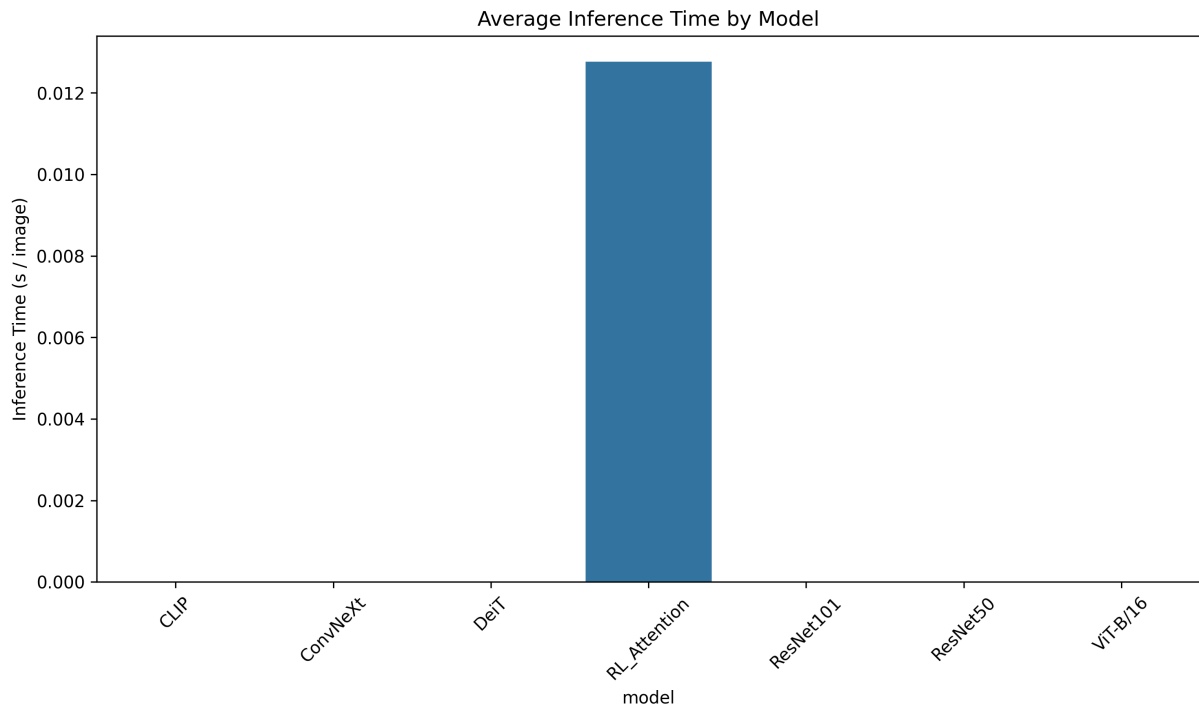


Figure 8: Average inference time comparison (seconds per image).

9. Conclusion & Next Steps

This empirical report confirms that active, recurrent visual foveation is a powerful strategy to overcome visual noise and cue conflicts. By shifting attention to shape-salient regions, the **RL Attention Agent** manages to bypass conflicting details that completely derail standard feedforward convolutional models.

Proposed Next Steps:

- Multimodal Evaluation:** Run the implemented [vlm_evaluation.py](#) script to benchmark state-of-the-art Visual Language Models (GPT-4o, Claude 3.5 Sonnet) on the same dataset to determine if large-scale instruction tuning bridges the shape-bias gap.
- Dynamic Glimpsing:** Implement an early-stopping saccade policy where the agent can choose to terminate glimpses early if its confidence threshold is met, reducing the inference time overhead shown in Figure 8.